WELLSLING · PROBLEM STATEMENT

WellSling problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WSL-PRB-001	Version	0.3
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/01-problem.md @ 346fa2d		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Users-and-needs and stakeholders tables added (kit 1.8 pipeline, Tables 2 and 3); user requirements now in WSL-URS-001
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 and 3; constraints restated for the kit's own tripod and the value-engineering target; open questions settled in WSL-DDR-001; first co-design candidates named; safety section added
0.1	2026-09-30	Amish Chadha	Initial scaffold

WellSling problem statement

Livestock fall into open wells in farming villages. People go down to get them out, and the bad air at the bottom kills them.

The problem

Wells and pits can hold air that is low in oxygen or full of gases, and a person can collapse within seconds of entry, as survivors of a NIOSH-investigated tank incident reported (NIOSH FACE 85-02). Farm families face this in pits and wells, and deaths come in chains: one person collapses and others follow (CDC MMWR, 1989; The Hawk, 2026). An animal in a well adds urgency and value, which pushes people to act before help arrives.

Existing non-entry equipment is built for people, not large animals. Commercial confined-space tripods carry a rated load of about 191 kg (420 lb) (3M) and cost several thousand dollars with a winch (PK Safety), while a grown cow or buffalo can weigh 400 to 600 kg (estimate). There is no low-cost, open way to get a lifting band under a heavy animal from the rim of a well.

Users and context

Table 1. Users and context.

User	Need	Context
Farmers and livestock owners	Get a fallen animal out without anyone climbing down	Open wells on farms and village edges, minutes after the fall
Village volunteers and dairy cooperative staff	A kit they can fetch, set up and use with a short briefing	Kit kept at a shared point in the village
Animal rescue organisations	A lighter, cheaper large-animal lift for well rescues	Rescue vans covering many villages
Rural fire and disaster response crews	A non-entry option for animal-in-well call-outs	Long travel to villages

Table 2. Users and needs.

User type	Goals and needs	Typical use scenario	Pain points and challenges
Farmers and livestock owners	Get a fallen cow, buffalo or calf out alive, quickly, without anyone going down	An animal falls into an open well on the farm at dusk; the family calls neighbours and fetches the village kit	The animal is worth a year's income; pressure to act before help arrives; bad air at the bottom has killed rescuers
Village volunteers and dairy cooperative staff	A kit they can fetch, set up and use after a short briefing, with clear steps	Three or four people carry the kit from the cooperative store and set it up over the well	Never used it before; rims crumble; no power; long, tiring lift by hand
Animal rescue organisations	A lighter, cheaper large-animal lift for well rescues that is gentle on the animal	A rescue van answers a call in a village 40 km away and works with the villagers	Existing confined-space tripods are rated for people; improvised rope lifts injure animals
Rural fire and disaster response crews	A non-entry method for animal-in-well calls that keeps their crew safe	Called out with the family already at the well and the animal in water	Long travel; their kit is for people; the urge to send someone down

Table 3. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Families of farmers and would-be rescuers	Bear the loss when a rescuer dies in a well	Awareness that nobody goes down; the kit kept and practised in the village
Dairy cooperative unions	Could buy, store and train with shared kits for member villages	Co-design partner candidate (named in Co-design); review of setup and training
Veterinarians and veterinary universities	Animal welfare during the lift: band width, pressure, time in the bands	Review of the bands and lift time before any live lift (WSL-DEC-001)
Animal rescue organisations	Run well rescues today; know real wells and animals	First co-design partner candidate; trial in a test pit at TRL 4
Village fabricators and workshops	Would make and repair the steel and timber parts	Check that every made part suits local tools and skills
Lifting equipment suppliers	Supply the winch, rope, shackles, chain and slings	Confirm catalogue data (winch handle force, sling eyes) when parts are bought
Local government and police	Called to accidents; may set rules for open wells	Informed of the kit; well covers and parapets remain the first prevention
Open-hardware community	Copies, improves and builds the design	Clear documents, open licences, no patent conflict

Operating environment

- Open, unlined or brick-lined wells assumed 1 to 3 m (3 to 10 ft) across and up to about 15 m (50 ft) deep for a first version (estimate).
- Rims may be crumbling, wet or unlined, with soft ground around them.
- Animal may be in water, standing, lying or wedged, and frightened.

- Hot, dusty or monsoon conditions; work often by day with no power on site.
- Air in the well may be low in oxygen or contain gases.

Constraints

- Value-engineering target USD 1,500 for the kit (a hypothetical control target, not a spending limit). The target was set when the kit was to borrow the HatchSide tripod; HatchSide is rated for people (about 191 kg class), so WellSling now carries its own 600 kg tripod and winch (WSL-DDR-001).
- Everything worked from the rim; no step may need a person to go down.
- Plain webbing belly band with no stiffening insert, and the passer as a separate tool (IP design-around).
- Operable by three to four untrained villagers after a short briefing.
- Open design: hardware under CERN-OHL-S-2.0.

Out of scope

- Rescue of people from wells.
- Veterinary care of the animal after lifting.
- Well covers, parapets and prevention, which are recommended but separate.
- Wells wider than 3 m, which need a different frame (for example an A-frame or a gantry over beams).

Prior work

Prior work	What it does	Gap for these users	Source
3M DBI-SALA confined space tripod	Aluminium tripod of 2.13 or 2.74 m for confined-space entry and rescue, rated 191 kg (420 lb)	Rated for people, not 400 to 600 kg animals, and costly	link
3M DBI-SALA tripod with Salalift II winch	Tripod and retrieval winch system with 18 m (60 ft) of cable, about USD 5,850 to 6,120	No means of passing a band under an animal; price out of reach for villages	link
US5431248A confined space lowering and retrieving apparatus	Expired patent for a tripod or quadruped frame over manways	Personnel system; free prior art for a frame	link

Co-design

A dairy cooperative or animal welfare organisation that already responds to animals in wells, ideally with a veterinarian, so the band, passer and lifting method are shaped around real animals, real wells and animal welfare during the lift. The first candidates to approach (none approached yet) are an animal rescue organisation that runs well rescues in north India, then a district dairy cooperative union in Haryana whose members lost people in the Jhajjar incident, with a state veterinary university for the band and handling advice (WSL-DDR-001, D8).

Questions settled at TRL 2

The scaffold's open questions were settled in WSL-DDR-001 under Amish's 2026-10-03 pre-approval:

- The shared HatchSide tripod cannot carry a 600 kg animal, so WellSling has its own tripod, rated at a 648 kg hook load with the line leaning up to 10 deg (D1); 662 kg after the 2026-10 fixes (WSL-CAL-001 v0.2).
- Wells wider than 3 m are out of scope for this kit (D2).
- An animal lying down or wedged is reached with the J-headed passer pole and a messenger line: the J goes under the body and the nose brings the line up the far side, and the line then pulls the band through (D4).
- Two plain webbing bands 240 mm wide, chest and flank, with loose felt sleeves, share the load; the veterinary view on pressure and duration is an item to confirm (D3).
- The first co-design candidates are named above (D8).

Safety

> **Safety:** The hazard that WellSling exists to remove is the bad air at the bottom of a well: nobody goes down at any stage, and if a person is already in the well, call the emergency services. The kit itself lifts a 600 kg animal over an open hole, so its own hazards are a falling or swinging load, a tipping tripod, a collapsing rim and a kicking animal. The design answers these with a load-brake winch, spread chains, pads set back from the rim, a landing bridge so the animal is never swung over the rim, and harnesses for anyone near the rim. It is an open engineering reference, not certified equipment.